QUESTION ONE

a) Explain any **five** ways by which computer viruses can be spread. [10]

b) A bank's ATM provides customers with three essential banking services: depositing money, withdrawing funds, and checking their account balance. Each customer begins with an initial balance of $500 in their account. Write a program that allows the user to perform multiple transactions through a loop and provides the following options:

- Deposit: Adds a specified amount to the account balance.
- Withdraw: Deducts a specified amount, but only if sufficient funds exist.
- Check Balance: Displays the current account balance.
- Ensures that a withdrawal cannot exceed the available balance, preventing overdrafts.
- Continues allowing transactions until the user chooses to exit

NB your program should provide a menu for the above options [15]

QUESTION TWO

a) Explain the functions of the following components in a computer system.

i. Arithmetic and Logic Unit (ALU). [2]

ii. Address Bus [2]

iii. Register [2]

 Copyright: National University of Science and Technology, 2025

b) An ICT manager at a University wants to upgrade the computer network that the institution is currently using. He has chosen the following network topology shown in the diagram below

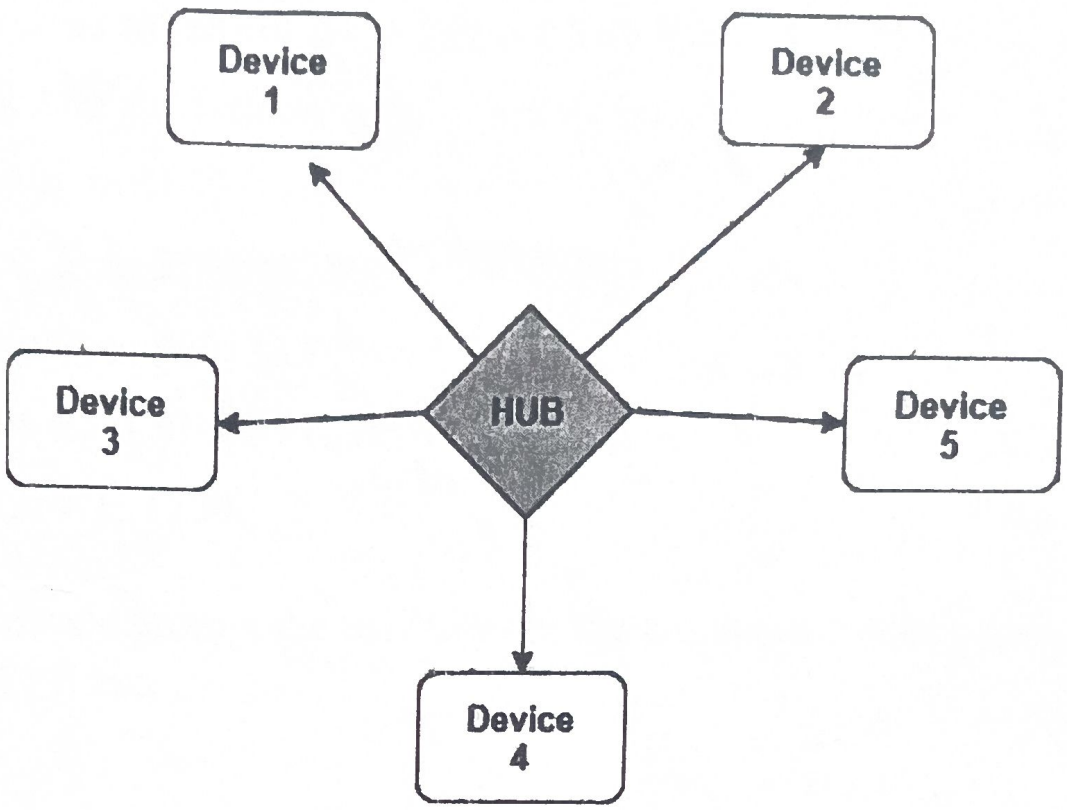


Figure 1

i. Identify the network topology that the ICT manager has chosen. [1]

ii. Outline **two** advantages and **one** disadvantages of using this topology. [3]

c) A Lecturer wants to analyze the exam scores of 30 students. She wants to enter all 30 scores into an array. Then, she needs to find the highest and lowest scores, calculate the average score, count how many students failed (score < 50). Write a C program that stores the exam scores in an array, prints the highest, lowest, and average scores, counts and prints how many students failed. [15]

int score [30] ;
Datatype arrayname [size] ;

QUESTION THREE

a) Define and provide an example for each of the following types of errors:

i. Syntax [2]
ii. Run time [2]
iii. Logical [2]
iv. Semantic [2]

#define PI 3.142
int x
int main ()

b) Variables can be local or global. Explain and illustrate with a program how you declare them in C. [7]

c) Compare and contrast Traditional filing system and cloud storage. [10]

int main () {
int x ;
x=17

Copyright: National University of Science and Technology, 2025

QUESTION FOUR

a) Explain any **five** benefits of using the internet. [10]

b) Write a C program to calculate the Pay As You Earn (PAYE) tax for an employee based on their salary. Use the following tax brackets for the calculation:

0% on income up to $100

20% on income from $101 to 300

25% on income from $301 to $1000

30% on income from $1001 to $2000

35% on income over $2000

The program should prompt the user to enter the employee's salary and then calculate and display the PAYE tax. [15]

QUESTION FIVE

a) Describe the following Cloud Computing models, and provide an example for each

i. IAAS [4]

ii. PAAS [4]

b) List any **five** functions of an operating system. [5]

c) Develop a completed C program that uses a recursive function to calculate the Fibonacci series. [12]

END OF QUESTION PAPER

 Copyright: National University of Science and Technology, 2025